... Lecture 6 ...
Single Locus Models

... Lecture 6 ...
Single Locus Models

Part 1: estimating effects of loci

Genetic Variation in Models

- Sources with 'random effect'
  - » infinitesimal model within breeds
  - » many loci within breeds i.e. approximate
  - » a set of inbred lines derived from a base
  - » a large sample of breeds (possibly)
- Sources with fixed effect
  - » differences between breeds or named lines
    - » unlikely to be a randomly sampled
  - » specific pre-determined loci

Initially we are looking at some 'fixed' genetic effects. Breeds are often tricky and one needs to question whether or not they are not a random sample of all breeds, but are present in the sample of breeds for special reasons e.g. they are commercially successful and hence extreme in some sense. Sometimes a breed or breed type will be highly related to others, such as a cross. Breeds are often artificially selected to a type, or naturally selected towards adaptation to native environment.

The focus in this lecture is when we examine the effect of a particular locus, which is the object of study for some *a priori* reason. (Look for analogies to QTL detection and analysis!)

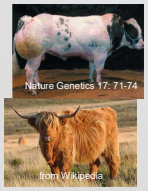
Identified causal variants in complex traits are commonplace. The scope and depth of genomic information is unprecedented.

Sequence Technology

- Have sequence level information on
  - » several 10^5 including within lines
  - » causal mutations discovered

Myostatin Gene

- Causes double muscling in cattle
- One of the first causative variants identified for a complex trait
  - » is it in other populations?
    - » YES!
  - » does it have the same effect?
    - » NO!
  - » is it pleiotropic?
    - » YES!



An example of a causal variant is the 'double muscling gene' within the GDF-8 locus, the myostatin locus. This was one of the first naturally occurring causal variants to be uncovered segregating within a species contributing variation to a complex trait (Grobet *et al*. 1997. Genetics, 17: 71-74). It has an advantageous effect on muscle mass, and carcass composition (typically additive effects), and has a deleterious impact on calving ease (with a recessive effect). It was identified in the Belgian Blue and Piedmontese breeds.

The discovery of such a causal variant affecting traits of interest will naturally raise questions: (i) "is the variant segregating in other populations?", (ii) "if so, is the magnitude and mode of action the same?", (iii) "does the variant have pleiotropic effects on traits other than those leading to its discovery?", and (iv) "are there other variants having a similar effect within the same locus?" In the case of myostatin the answer to these questions were: yes – South Devon and Highland also had the mutation segregating; yes – it has pleiotropic effects (muscling and calving ease); no – there was no obvious impact in Highland, which is odd; yes – there are other variants with effect.

Test Effects

- Natural experiment to look at these questions
  - » sample population
  - » measure traits of interest
  - » genotype the locus
  - » fit a model

29 January 2022 Lecture 6

However, the first step is to use the natural experiment of a field study, by sampling the population, often at random, genotype individuals and measure the traits of interest, and then fit a model.

Least Squares Model

- Model $\mathbf{y} = \mathbf{1}m + \mathbf{X}_1\mathbf{b} + \mathbf{X}_2\mathbf{g} + \mathbf{e}$
  - » **y** vector of observations (n x 1)
  - » **1***m* *m* is mean effect, fixed (1 d.f.)
    - » design vector **1** is (n x 1) vector of 1's
  - » **b** nuisance effects, (fixed) with design $\mathbf{X}_1$
  - » **g** QTL locus, fixed with design $\mathbf{X}_2$ (2 d.f.)
  - » **e** (n x 1) vector of residual effects
    - » **e** random, assumed MVN(0,σ_E^2**I**)

29 January 2022 Lecture 6

It would seem superficially that a suitable model might be to fit the nuisance factors (e.g. sex, farms, some of which may be considered to be random) and to fit the QTL locus as it is known and considered to confer systematic effects on the phenotype. In the slide I have just fixed effects for the nuisance factors – but we know how to write more complex mixed models.

However this model would be wrong!!!

Complex Traits

- For a complex trait one must expect multiple loci with effects
- If this is neglected, only way to explain genetic change is the identified locus
  - » overestimate the effect
  - » particularly if change is caused by selection

29 January 2022 Lecture 6

The reason is that for complex traits one must account for all the remaining genetic variation in all the traits of interest that are not accounted for by the causal variant. Otherwise the model explains genetic structure and change, e.g. through selection, solely through the locus of interest.

Polygenic Variance

- Need to account for other genetic variation
  - » plus any selection
- Introduce a polygenic effect
  - » typically assumed to be 'polygenic'
    - » many alleles of small effect
  - » modelled by random effects
- Paper showing bias and polygenic correction
  - » Kennedy, Quinton & van Arendonk
  - » J. Anim. Sci. 1992, 70: 2000-2012

29 January 2022 Lecture 6

This is called 'polygenic' variation and is typically assumed to be made of many alleles of small effect.

The paper that set out and investigated the problems and solutions for estimating the effect of a single locus in complex traits was Kennedy, Quinton and van Arendonk (1992).

Polygenic Model

- Model $\mathbf{y} = \mathbf{1}m + \mathbf{X}_1\mathbf{b} + \mathbf{X}_2\mathbf{g} + \mathbf{Z}\mathbf{u} + \mathbf{e}$
  - » **y** vector of observations (*n* x1)
  - » **1***m* *m* is mean effect, fixed (1 d.f.)
    - » design vector **1** is (*n* x1) vector of 1's
  - » **b** nuisance effects, fixed with design $\mathbf{X}_1$
  - » **g** QTL locus, fixed (2 d.f.) with design $\mathbf{X}_2$
  - » **u** (*p* x1) vector of polygenic breeding values
    - » design matrix **Z**, **u** random MVN(0,σ_A^2**A**)
  - » **e** (*n* x1) vector of residual effects,
    - » **e** assumed random MVN(0,σ_E^2**I**)

31 January 2022 Lecture 6

Therefore, the correct model is a linear mixed model (mixed as it contains both fixed *and* random effects). The QTL may still be fitted as a fixed effects and the polygenic breeding values are modelled as random effects, with variance modelled by the Numerator Relationship Matrix, i.e. MVN(0, σ_A^2**A**). With genomic data this might be replaced with a genomic relationship matrix, which we will meet next week.

Here *n* is the number of observations being analysed in the **y**, and *p* is the number of individuals represented in the relationship matrix (the numbers in pedigree and/or genotyped)

Case 1: No Effect of Locus

- Ignoring polygenic effects gives too many significant effects
  - » ↑ Type I error
  - » problems increase with selection
  - » due to poor estimate of σ_e^2 through not accounting for relationships when $\sigma_A^2 > 0$
- Model with polygenic term shows no increase in Type I error

29 January 2022 Lecture 6

Remember Type I error is 'accepting Alternative Hypothesis when Null Hypothesis is true'. (Type II error is 'accepting Null Hypothesis when Alternative Hypothesis is true').

The Type 1 error is increased if the genetic sub-structure (e.g. families) is ignored even if *there is no effect of the locus and no selection.* Neglecting genetic relationships will underestimate the residual variance – this alone will increase Type 1 error. Selection will compound this effect as the selection response becomes associated with allele frequency change over time, either through random drift at the locus or through linkage to true QTL. When a polygenic effect is fitted the control of Type 1 error is restored.

Case 2: Locus With Effect

- Without polygenes, with selection
  - » *a* biased upward across generation analyses
    - » all selection response being explained by QTL!
  - » *a* biased downward within generation analyses
    - » due to negative linkage disequilibrium
    - » an animal with poor QTL alleles tends to have good polygenes and vice versa
  - » *d* has less bias
- Model with polygenic term is unbiased

29 January 2022 Lecture 6

When the locus does have an effect, and there is selection, then ignoring the genetic sub-structure will bias the estimated effect upwards when data spans multiple generations; it is the only term that can explain the genetic trend evident in the data. However, if analyses are made within generations, the bias is downwards because of the negative linkage disequilibrium generated by the selection.

This negative covariance can be hard to imagine. It can be shown algebraically, but may been seen from the following heuristic argument. If an individual has been selected as being of superior genetic merit, and one is told that the alleles carried at one of the loci are poorer than average, then one might infer that, pooled over the remaining loci, the alleles are likely to be better than average since the individual has superior genetic merit i.e. a negative covariance!

If there is an effect but no selection, ignoring the genetic structure will underestimate the sampling error of the estimate.

Conclusion

- Models must account for polygenic variation in complex traits
  - » control Type I error
  - » avoid bias in estimates, particularly *a*
- Bias without polygenic term increase with
  - » higher selection intensity
  - » number of generations
  - » i.e. total selection differential applied

Reference Kennedy, Quinton & Van Arendonk, 1992

29 January 2022 Lecture 6

Therefore, when estimating the effect of a pre-determined locus as a fixed effect the effects associated with the remaining genetic variance must be modelled. This can be done by fitting polygenic breeding values, assumed to have a (co)variance matrix described by **A** (pedigree models) or a genomic relationship matrix (genomic models). This will remove bias in the estimate and give the desired Type 1 error rate.

Fitting Model

- Use REML because polygenic σ_A^2 unknown
- Use Wald-tests (or t-tests for 1 d.f.).
  - » t-test is 'estimate'/'estimated s.e.' for Normal
  - » Wald tests generalise this and use $\hat{\mathbf{x}}^T\mathbf{V}^{-1}\hat{\mathbf{x}}$
  - » always approximate, but asymptotically OK
  - » ASReml does it for you!
- Use 'qgm_contrast()' to estimate contrasts

29 January 2022 Lecture 6

These models can be fitted using ASReml, using Wald tests to test for the effects of the pre-determined locus fitted as fixed effects. In ASReml-R we can use `qgm_contrast()` to estimate relevant parameters associated with the locus.

Genetic Interpretation

- For an A/T polymorphism
- Fitting 2 d.f. for QTL
  - » test $\mathbf{H_0}$: $g_{AA} = g_{AT} = g_{TT}$ vs $\mathbf{H_1}$: not H_0
  - » Wald test of genotype means

29 January 2022 Lecture 6

The Wald test hypotheses!

Genetic Interpretation

- Estimate
  - » $a = ½ (g_{TT} - g_{AA})$ 'additive effect'
    - » contrast is (-0.5, 0.0, 0.5) of means for g_{AA}, g_{AT}, g_{TT}
  - » $d = g_{AT} - ½ (g_{AA} + g_{TT})$ 'dominance deviation'
    - » contrast is (-0.5, 1.0, -0.5) 'effect of allelic substitution' by T
  - » $\alpha_T = a + (p_A - p_T)\, d$ for frequency p_T and p_A=1-p_T
    - » contrast is $(-p_A, (p_A - p_T), p_T)$
  - » $u_T = (1-p_T)\, \alpha_T$, contrast is $(-p_A^2, p_A(p_A-p_T), p_A p_T)$
  - » $u_A = -p_T \alpha_T$, contrast is $(p_A p_T, -p_T(p_A-p_T), -p_T^2)$ 'average effects' of T and A

29 January 2022 Lecture 6

All these are standard effects introduced in QG and described in Falconer & Mackay. They are: (i) the additive effect, half the difference between the homozygotes; (ii) the dominance effect, the difference between the heterozygote and the average of the homozygotes; (iii) the effect of an allelic substitution (T in this A/T example) and (iv) the average effects. The average effects are the allelic effects scaled so they have mean 0, i.e. $p_T u_T + p_A u_A = 0$ and are derived from the allelic substitution effect. Note the effect of an allelic substitution and the average effects are functions of allele frequency.

Standard Errors

- Need (co)variance matrix **V** of estimates
  $\mathbf{g}^T = (g_{AA}, g_{AT}, g_{TT})$
- Then e.g. additive effect
  - » $a = \mathbf{c}^T\mathbf{g}$ where $\mathbf{c}^T$ = (-0.5, 0.0, 0.5)
  - » $var(\mathbf{c}^T\mathbf{g}) = \mathbf{c}^T\mathbf{V}\,\mathbf{c}$ and s.e. = $\sqrt{\mathbf{c}^T\mathbf{V}\mathbf{c}}$
    - » see Merry Matrices
- All done by 'qgm_contrast()'!

29 January 2022 Lecture 6

All these are contrasts of the genotype means, and the analysis gives us estimates of these contrasts. Remember a contrast is a set of coefficients for the genotype means that sum to 0; e.g. the dominance deviation is described by the contrast **c** = (-½, 1, -½).

For a contrast, we can estimate its sampling variance by $\mathbf{c}^T\mathbf{V}\mathbf{c}$, where **V** is the sampling (co)variance matrix of the genotype means calculated by ASReml. The s.e. is then the square root, i.e. $(\mathbf{c}^T\mathbf{V}\mathbf{c})^{1/2}$. Or we can use `qgm_contrast()`.

Model

- Model $\mathbf{y} = \mathbf{1}m + \mathbf{X}_1\mathbf{b} + \mathbf{X}_2\mathbf{g} + \mathbf{Z}\mathbf{u} + \mathbf{e}$
  - » **y** vector of observations (n x 1)
  - » **1**m m is mean effect, fixed (1 d.f.)
    - » design vector 1 [illegible] vector of 1's
  - » **b** nuisance effects, fixed with design $\mathbf{X}_1$
  - » **g** QTL locus, fixed (2 d.f.) with design $\mathbf{X}_2$
  - » **u** (p x 1) vector of polygenic breeding values
    - » design matrix **Z**, **u** assumed random MVN(0,σ_A^2**A**)
  - » **e** (n x 1) vector of residual effects
    - » **e** assumed random MVN(0,σ_E^2I)

What is the breeding value?

29 January 2022 Lecture 6

Breeding Value

- Additive genetic merit (breeding value) for i with population frequencies p_A & p_T
  - » with genotype AA is $u_i + 2u_A$
  - » with genotype AT is $u_i + u_A + u_T$
  - » with genotype TT is $u_i + 2u_T$

29 January 2022 Lecture 6

The contribution to the breeding values from the identified locus are the sum of the average effects of the 2 alleles carried by an individual, e.g. a heterozygote AT has breeding values u_T+u_A. These are also contrasts of the fixed effects and functions of the allele frequencies.

The full breeding value is the sum of the estimate from the polygenic term and the identified locus.

Single Locus Genotype

- Treated QTL as fixed
  - » estimate a, d, allelic substitutions
- If use 'least squares' fixed model
  - » increased Type I error
  - » bias with selection, particularly a is overestimated
- With linear mixed model
  - » fit 'polygenic' variance, with σ_A^2**A**
  - » bias removed, Type I error controlled
  - » may need MCMC if missing genotypes

29 January 2022 Lecture 6

The summary …

Application

- Gene Assisted Selection
  - » e.g. double muscling locus
  - » use BLUP to estimate polygenic EBV, add locus effects
- OK, <u>providing</u>
  - » it is actually causative and not a linked locus
    - » otherwise LD will break down!
  - » gene effects are very accurately estimated
    - » otherwise lose accuracy and lose gain

29 January 2022 Lecture 6

This model can be applied in prediction, e.g. in selection and in a selection context, it is termed Gene Assisted Selection (GAS). It is used for causal variants, and one should have data that is sufficient to estimate the genotype means with only small sampling error compared to the effect size and the polygenic genetic s.d.

The use of causal variants is carried out for IPN (Infectious Pancreatic Necrosis) resistance in salmon breeding, for or against the double muscling allele, and in risk prediction in medicine. As data sets increase in record numbers it is probable they will change to a more Bayesian perspective, i.e. including these loci in the predictive model from the SNP arrays which we will discuss next week.

Note that the genetic variance explained by a locus, which is of interest, does not really explain the value of a QTL to a population, or its utility for selection. For selection a favourable allele at a low frequency is very useful, but the favourable allele at a high frequency is not! The genetic variance explained at the locus is the same in both cases as it depends on $p_T(1-p_T)$. *However*, the average effects do describe the utility – the value is $(1-p_T)\alpha_T$, so if p_T is high, the average effect is low, and if p_T is low, the average effect is high!

... Lecture 6 ...
Single Locus Models

Part 2: using markers to track IBD

29 January 2022 Lecture 6

Back to Genetic Variance

- How much variance is explained by QTL?
- In principle, $2pq\alpha_T^2$, however α_T^2 is not estimated in analysis … only α_T is estimated
  - » but $E[\hat{\alpha}_T^2] = E[(\alpha_T + \varepsilon)^2] = \alpha_T^2 + E[\varepsilon^2] = \alpha_T^2 + \mathrm{var}(\varepsilon)$
- Strict square $\hat{\alpha}^2$ will overestimate variance
  - » by an amount depending on estimation error
  - » more said in LAGA

29 January 2022 Lecture 6

Nevertheless, it is a natural question to ask is how much of the genetic variance is explained by the locus. This will depend on $2pq\alpha_T^2$. However, from the fixed effect model we only have an estimate for α_T, not α_T^2, and this has implications.

$$E[\widehat{\alpha_T}^2] = E[(\alpha_T + \epsilon)^2] = \alpha_T^2 + E[\epsilon^2] = \alpha_T^2 + var(\epsilon) \;>\; \alpha_T^2 ,$$

Therefore we should expect the estimate of α_T squared to be greater than the true value squared by an amount equal to its sampling variance. Therefore the genetic variance will be overestimated by using squaring the estimate. Note there is no covariance between α_T and its sampling error … if there was we could use it to improve the estimate!

Issues

- How much variance does locus explain?
- What happens if the QTL is not causative but linked?

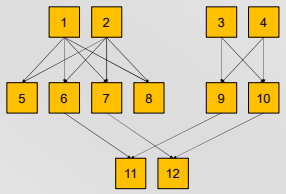
Revision on A

- **A** is the numerator relationship matrix between 2 individuals defined by

  $\mathrm{cov}(g_u, g_v) = a_{uv}\sigma_A^2$

  - » g_u, g_v are breeding values
  - » where σ_A^2 is variance in base population
  - » shown to be twice coefficient of kinship, $2\varphi_{uv}$
  - » dimensions *n* x *n* where *n* is # individuals

Wright's Numerator Relationship Matrix (**A**) is concerned with the covariance matrix of breeding values, which is seen to be twice the coefficient of kinship. Therefore, it is tracking probabilities of IBD defined from a base generation. It is dimension *n* x *n*, where *n* is the number of individuals in the pedigree.



A pedigree of 12 individuals …

A-matrix

	1	2	3	4	5	6	7	8	9	10	11	12
1	1.000											
2	.	1.000										
3	.	.	1.000									
4	.	.	.	1.000								
5	0.500	0.500	.	.	1.000							
6	0.500	0.500	.	.	0.500	1.000						
7	0.500	0.500	.	.	0.500	0.500	1.000					
8	0.500	0.500	.	.	0.500	0.500	0.500	1.000				
9	.	.	0.500	0.500	.	.	.	.	1.000			
10	.	.	0.500	0.500	.	.	.	.	0.500	1.000		
11	0.250	0.250	0.250	0.250	0.250	0.500	0.250	0.250	0.500	0.250	1.000	
12	0.250	0.250	0.250	0.250	0.250	0.250	0.500	0.250	0.250	0.500	0.250	1.000

… with A-matrix, where '.' means 0.

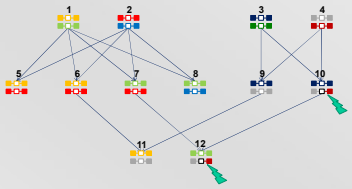
A-matrix at (say) 12 cM

	1	2	3	4	5	6	7	8	9	10	11	12
1	1.000											
2	.	1.000										
3	.	.	1.000									
4	.	.	.	1.000								
5	0.500	0.500	.	.	1.000							
6	0.500	0.500	.	.	0.500	1.000						
7	0.500	0.500	.	.	0.500	0.500	1.000					
8	0.500	0.500	.	.	0.500	0.500	0.500	1.000				
9	.	.	0.500	0.500	.	.	.	.	1.000			
10	.	.	0.500	0.500	.	.	.	.	0.500	1.000		
11	0.250	0.250	0.250	0.250	0.250	0.500	0.250	0.250	0.500	0.250	1.000	
12	0.250	0.250	0.250	0.250	0.250	0.250	0.500	0.250	0.250	0.500	0.250	1.000

A is based on IBD, applies to a single locus!

Consider a chromosome with markers which is ungenotyped and contains a locus of interest.

The relationship matrix for a locus at 12 cM based on the pedigree is simply Wright's Numerator Relationship Matrix – assuming it is neutral and selection-free – as it is for any other position; if not neutral and selection-free there are very few options for assuming anything else!



Assume locus of interest at (say) 12 cM
- flanking markers can be fully phased
- markers are sufficiently close to ignore double recombinations

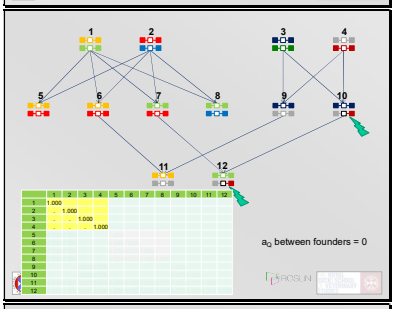
Now assume that genotypes are available at flanking markers and are sufficiently informative enough to distinguish the 8 base generation DNA fragments at this location, sufficiently close to ignore double recombinations. This is what technology offers.

A_Q: Locus Specific A

- Let Q be the ungenotyped locus at 12 cM
- Can we calculate $\mathbf{A}_Q$
  - » use markers to trace inheritance back to base
  - » calculate probabilities of IBD
  - » i.e. φ_{uv} … kinship between u and v at 12 cM
  - » so $a_{uv} = 2\varphi_{uv}$ at 12 cM

29 January 2022 Lecture 6

We will calculate the relationship matrix for an unidentified locus at 12 cM by using the markers to trace inheritance back to the base. This will be achieved by calculating kinships for each pair at 12 cM, and then use $a_{uv}=2\phi_{uv}$



The base is easy! Same as standard **A**.

A_Q: Locus Specific A

- If two individuals have no common ancestor
  - » $a_{uv} = 0$, at all positions
  - » so if $a_{uv} = 0$ in Wright's **A**, then also in $\mathbf{A}_Q$

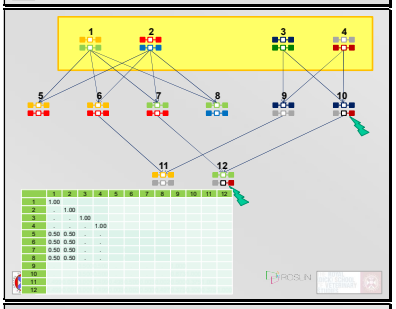
29 January 2022 Lecture 6

A_Q: Locus Specific A

- Parent offspring with parent in base
  - » $\varphi_{uv} = 0.25$, $a_{uv} = 2\varphi_{uv} = 0.5$
  - » For example 1 & 5 …
    - » need to sample 1's gamete from 6: P = 0.50
    - » need same at 12 cM from 1: P = 0.50
    - » $\varphi_{uv} = 0.5 \times 0.5 = 0.25$
    - » $a_{uv} = 2\varphi_{uv} = 0.5$
  - » in this case markers don't change a_{uv}

29 January 2022 Lecture 6

If say u is an offspring of v then ϕ_{uv} =0.25 and a_{uv}=0.5, otherwise 0 as we are considering IBD with respect to the base.



A_Q: Locus Specific A

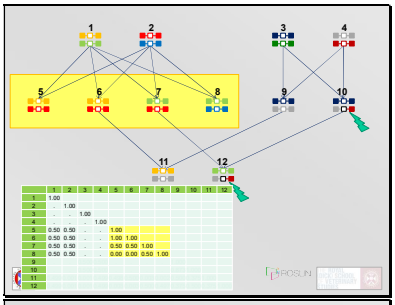
- $\varphi_{5,6} = 0.50$, $a_{5,6} = 1.0$
  - » can ignore double recombination
- $\varphi_{5,7} = 0.25$, $a_{5,7} = 0.5$
- $\varphi_{5,8} = 0.00$, $a_{5,8} = 0.0$
- …

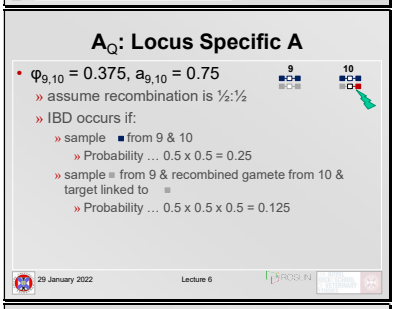
29 January 2022 Lecture 6

When it comes to individuals 5 to 8 then things are modified by what we observe in the markers. We are assuming the double recombination can be ignored and so we can calculate ϕ_{uv} by considering the sampling with replacement as demonstrated in QG.

None of individuals 5 to 8 have IBD markers so their ecoefficients with themselves is ϕ_{uu}=1; if they were IBD then ϕ_{uu}=1 and would be a_{uu}=2.

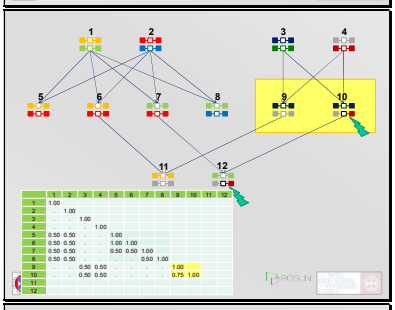
Among pairs of individuals (u,v), it is clear that 5 and 8 have no IBD relationships among the alleles so $\phi_{5,8}$ and $a_{5,8}$=0. For 5 and 6, $\phi_{5,6}$=0.5 and $a_{5,8}$=1. The markers have indicated a change to Wright's matrix as a result of observing the allele sampling that has taken place.



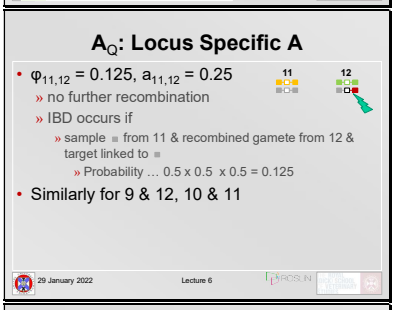


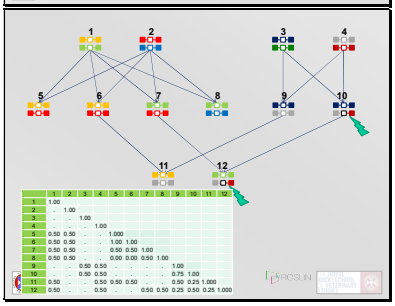
Sometimes a recombination will take place within the fragment of DNA being used to track the markers, as here in individual 10. For the purposes of this lecture we will assume that the true locus of interest has ½ chance of being recombined with either of the recombining fragments– it could be weighted differently if we could identify the exact position of the crossover.

Considering $\phi_{9,10}$ the average can be taken of the ϕ at either end of the interval, and, in this example $\phi_{9,10}$=0.375 and $a_{9,10}$=0.75.



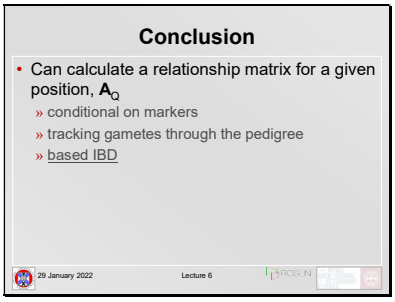
And so onwards down the pedigree, looking at the markers inherited by the offspring and relating these to the markers carried by the parents, and so calculating ϕ and hence a.





At the end we arrive at a relationship matrix $\mathbf{A}_Q$ for an unidentified locus, known only by its location on the chromosome, can be derived by using dense flanking markers to track the Mendelian sampling of the gametes at that position and hence calculating ϕ_{uv} and hence a_{uv}.

To carry this out with a full pedigree over multiple generations one would need to introduce what is termed the *Gametic Relationship Matrix* which is a 2n x 2n matrix relating all gametes to all other gametes among the *n* individuals base generation. However QGM cannot do everything!



The take-home message from this part of the lecture is that the markers allow us to track the actual Mendelian sampling from parent to offspring at a position on the genome without having to say probability ½ the paternal allele and probability ½ the maternal allele.

Three points to note: (i) it is based on IBD, not IBS; (ii) therefore if two individuals are unrelated in **A**, they remain unrelated in $\mathbf{A}_Q$; and (iii) the segregation probabilities are now informed by markers not assuming ½:½ and so account for selection pressures that may be applied to that fragment.

... Lecture 6 ...
Single Locus Models

Part 3: estimating variances using a locus specific $\mathbf{A}_Q$

Using $\mathbf{A}_Q$

- Can use $\mathbf{A}_Q$ in linear mixed model to calculate the effect of the locus Q
- However must include polygenic **A** to capture other variation

This gives us the Marker Assisted Selection (MAS) model, which produces Marker Assisted BLUPs (MA-BLUP) for estimates of breeding value (genetic merit). Just as when fitting a single locus as a fixed effect, all the other loci potentially affecting the complex trait need to be accounted for, and a polygenic term is required as before. However now the locus is being fitted as a random effect to estimate the variance explained at that position on the genome.

MAS Model

- Model $\mathbf{y} = \mathbf{1}m + \mathbf{X}\mathbf{b} + \mathbf{W}_1\mathbf{g} + \mathbf{W}_2\mathbf{g}_Q + \mathbf{e}$
  - » **y** (nx1) vector of observations
  - » **1**m mean effect m, fixed (1 d.f.) with design **1**
  - » **b** nuisance effects, fixed with design **X**
  - » **g** (px1) vector of BV, random with design $\mathbf{W}_1$
    - » **g** assumed MVN(0,$\sigma_A^2\mathbf{A}$)
  - » $\mathbf{g}_Q$ (px1) vector Q-BV, random with design $\mathbf{W}_2$
    - » $\mathbf{g}_Q$ assumed MVN(0, $\sigma_Q^2\mathbf{A}_Q$)
  - » **e** (nx1) residual effects, random
    - » **e** assumed MVN(0,$\sigma_E^2\mathbf{I}$)

This is the MAS model. We know how to write models equations! Note the modification from the fixed model. The MAS model uses (imprecise) knowledge of a QTL for prediction of genetic merit and breeding values.

Note that in the MAS model the locus is modelled as a MVN() – there are not just 3 genotype values! The 'locus' is perhaps best viewed as a fragment with lots of haplotypes spanning the fragment.

Fitting MAS Model

- Fitting MAS Model will estimate σ_Q^2 and $\hat{\mathbf{g}}_{Qu}$
  - » EBV of u is $\hat{\mathbf{g}}_u + \hat{\mathbf{g}}_{Qu}$
  - » QTL variance is σ_Q^2
  - » total genetic variance is $\sigma_A^2 + \sigma_Q^2$
  - » phenotypic variance $\sigma_P^2 = \sigma_A^2 + \sigma_Q^2 + \sigma_E^2$
  - » 'total' $h^2 = (\sigma_A^2 + \sigma_Q^2)/\sigma_P^2$
  - » 'polygenic' $h_A^2 = \sigma_A^2/\sigma_P^2$
  - » 'QTL' $h_Q^2 = \sigma_Q^2/\sigma_P^2$

The EBVs and parameters can be calculated. Now each individual has 2 EBVs, one for polygenic and one for the QTL. In this model the total genetic variance is $\sigma_A^2+\sigma_Q^2$. One can consider 3 heritabilities defined by σ_A^2 only, σ_Q^2 only, and their sum $\sigma_A^2+\sigma_Q^2$, all as fractions of the phenotypic variance $\sigma_A^2+\sigma_Q^2+\sigma_E^2$. The EBV of an individual *u* would be the sum of the polygenic and QTL EBVs.

Linkage Analysis

- This is linkage analysis (LA)
  - » calculates IBD using markers
  - » relies on having pedigree
  - » uses markers to track inheritance down pedigrees
    - » within families initially
    - » then distributes updated information to descendants
  - » 'pattern' similar to **A**
    - » has 0's where Wright's NRM is 0
  - » doesn't track population-wide IBS
    - » does not make assumptions on IBS in base generation

This general approach is often referred to as linkage analysis.

… Lecture 6 …
Single Locus Models

Part 4: concluding notes

Linkage Analysis

- LA historically used for QTL mapping
  - » now less common with introduction of GWAS
- Models are 'Marker Assisted BLUP'
  - » foundation of Marker Assisted Selection or MAS
    - » Fernando & Grossman, Genetics Selection Evolution, 1989, 21: 467-477
  - » contrast with GAS with known causative marker
    - » and fitted as fixed

29 January 2022 Lecture 6

Linkage analysis was used historically for QTL mapping as well as prediction, typically with micro-satellites before the introduction of really dense SNP panels.

The key publication which opened up these linear mixed models with single position relationship matrices was Fernando and Grossman (1989).

Linkage Analysis

- LA historically used for QTL mapping
  - » now less common with introduction of GWAS
- Models are 'Marker Assisted BLUP'
  - » foundation of Marker Assisted Selection or MAS
    - » Fernando & Grossman, Genetics Selection Evolution, 1989, 21: 467-477
  - » contrast with GAS with known causative marker
    - » and fitted as fixed

Neither GAS nor MAS were very successful!

29 January 2022 Lecture 6

The need for theses MAS models was prompted by the fact that early QTL detection methods and bioinformatic knowledge could not localise QTLs very closely, and consequently estimates of the QTL effects were very error prone (even if the QTL was not a false positive!). The use of Marker-Assisted BLUP reduced some of the risks from using the QTL information as a fixed effect; when fitted as random effects the effects are shrunk in relation to the information available.

MAS was not very successful: the models were computationally demanding (at that time) as every 'QTL' included, and the polygenic term, need to have its variance component estimated relatively precisely and then fitted in the model. Obtaining reliable estimates also demanded sizeable datasets. Unreliable parameters can reduce efficiency compared to ignoring the QTL!).

GAS & MAS

- GAS & MAS have a 'QTL perspective'
  - » assume can explain variation by finding a handful of QTLs
  - » can neglect other genetic variance for prediction

29 January 2022 Lecture 6

One reason for the lack of success of GAS and MAS in predicting genetic merit (breeding values) in complex traits is that they are based on a QTL perspective – that genetic variation can be satisfactorily accounted for, and accurate predictions of breeding value can be made, by observing just a handful (or a few handful) of QTLs.

This is not the case for complex traits: a different approach is required!

Widening the Perspective

- With dense markers can calculate $\mathbf{A}_Q$ for many positions across the genome!
- Average $\mathbf{A}_Q$ to get a genomic relationship matrix over the whole genome, say $\mathbf{G}_{FG}$
- Based on IBD!
- Could use for evaluations in place of **A**
- A whole-genome genomic relationship matrix!

29 January 2022 Lecture 6

However we will return to $\mathbf{A}_Q$ briefly in future lectures. One could calculate $\mathbf{A}_Q$ for many, many positions in the genome and average them. This gives a genomic relationship matrix: using markers to infer IBD, directly linked to the pedigree which may also have been inferred generation by generation using the markers. This could be used in evaluations in place of **A**, and could be used in optimum contribution methods (remember QG) to control inbreeding and in evaluations.

However this requires a pedigree, often not available or impossible to construct, so again a different approach may be desirable!